

CSC577

SOFTWARE ENGINEERING: THEORIES AND PRINCIPLES

LAB: 021 - PROJECT MANAGEMENT



اَوَّلُ سَبِيلٍ سَبِيلُ الْمَعْرِفَةِ
UNIVERSITI
TEKNOLOGI
MARA

SOFTWARE PROJECT MANAGEMENT DISTINCTIONS

- The product is intangible
 - The product is uniquely flexible
 - Software engineering is not recognized as an engineering discipline with the same status as mechanical, electrical engineering, etc
 - The software development process is not standardized
 - Many software projects are 'one-off' projects

SOFTWARE PROJECT MANAGEMENT

- Ensuring that software is delivered
 - on time
 - on budget
 - in accordance with the requirements
- Project management is needed because software development is always subject to budget and schedule constraints that are set by the organisation developing the software



WHY PROJECT MANAGEMENT?

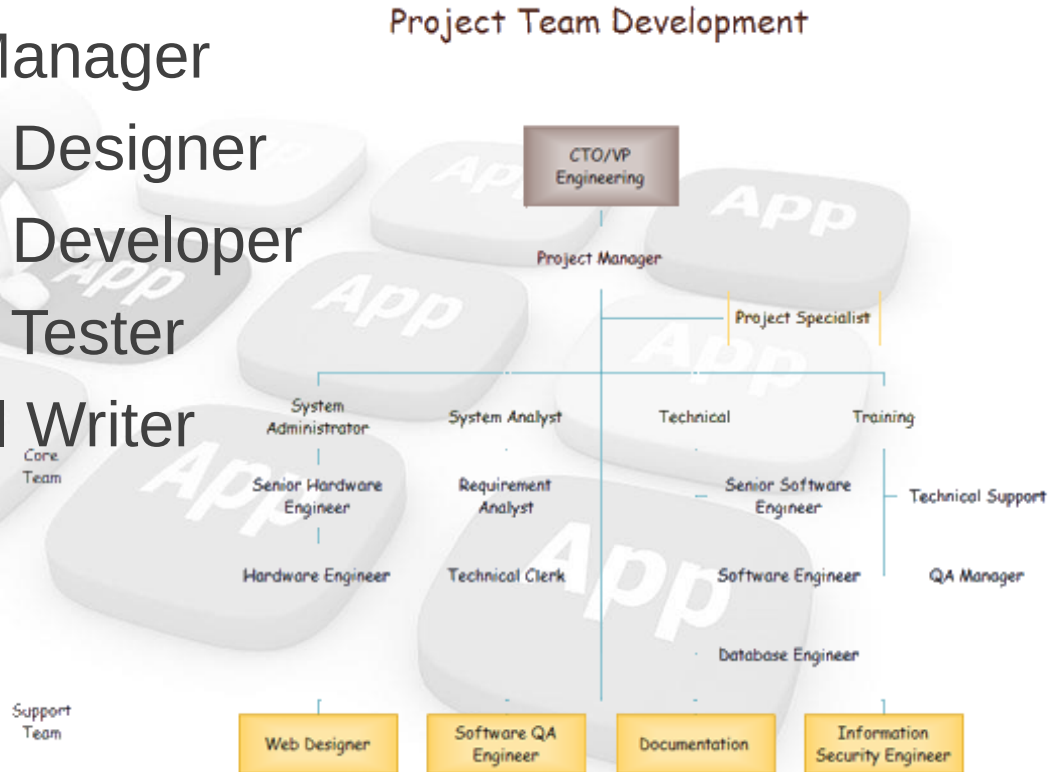
- Good PM is not enough to warranty that a project succeeds
 - But bad PM is enough to warranty that a project is late, over budget and does not deliver the needed functionality
- Key activities in PM are project planning, cost and effort estimation, project tracking, project organization, risk management
- Key tools are Work breakdown structure, product breakdown structure (VPM in software projects), Gantt and Pert charts, process and product measures

PROJECT STAFFING 1/2

- May not be possible to appoint the ideal people to work on a project
 - Project budget may not allow for the use of highly-paid staff
 - Staff with the appropriate experience may not be available
 - An organisation may wish to develop employee skills on a software project
- Managers have to work within these constraints especially when there are shortages of trained staff

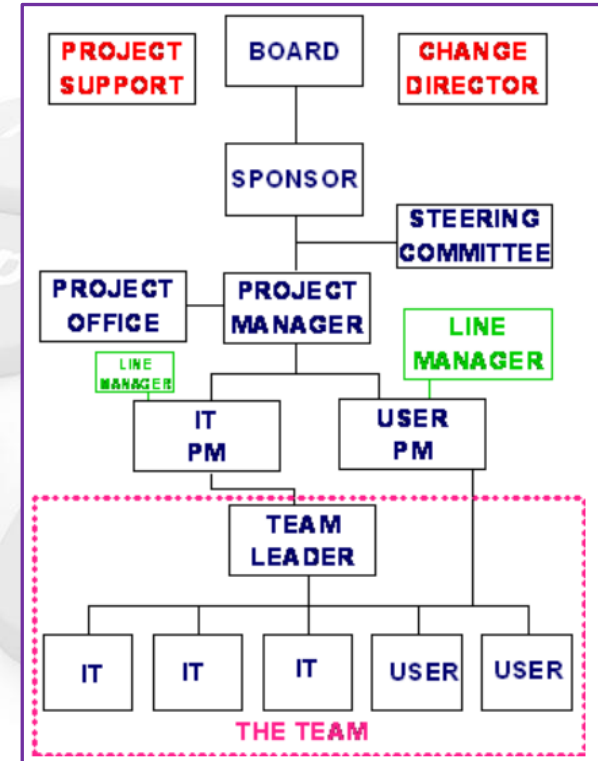
PROJECT STAFFING 2/2

- Project Manager
- Software Designer
- Software Developer
- Software Tester
- Technical Writer



PROJECT ROLES & RESPONSIBILITIES

- Project Sponsor
- Project Steering Committee (Project Board)
- Project Director
- Project Manager
- Project Office
- Project IT Manager
- Project User Manager
- Project Leader
- Team Leader
- Quality Leader
- Team Member
- Project Assurance
- Project Support
- Internal Audit
- Line Manager



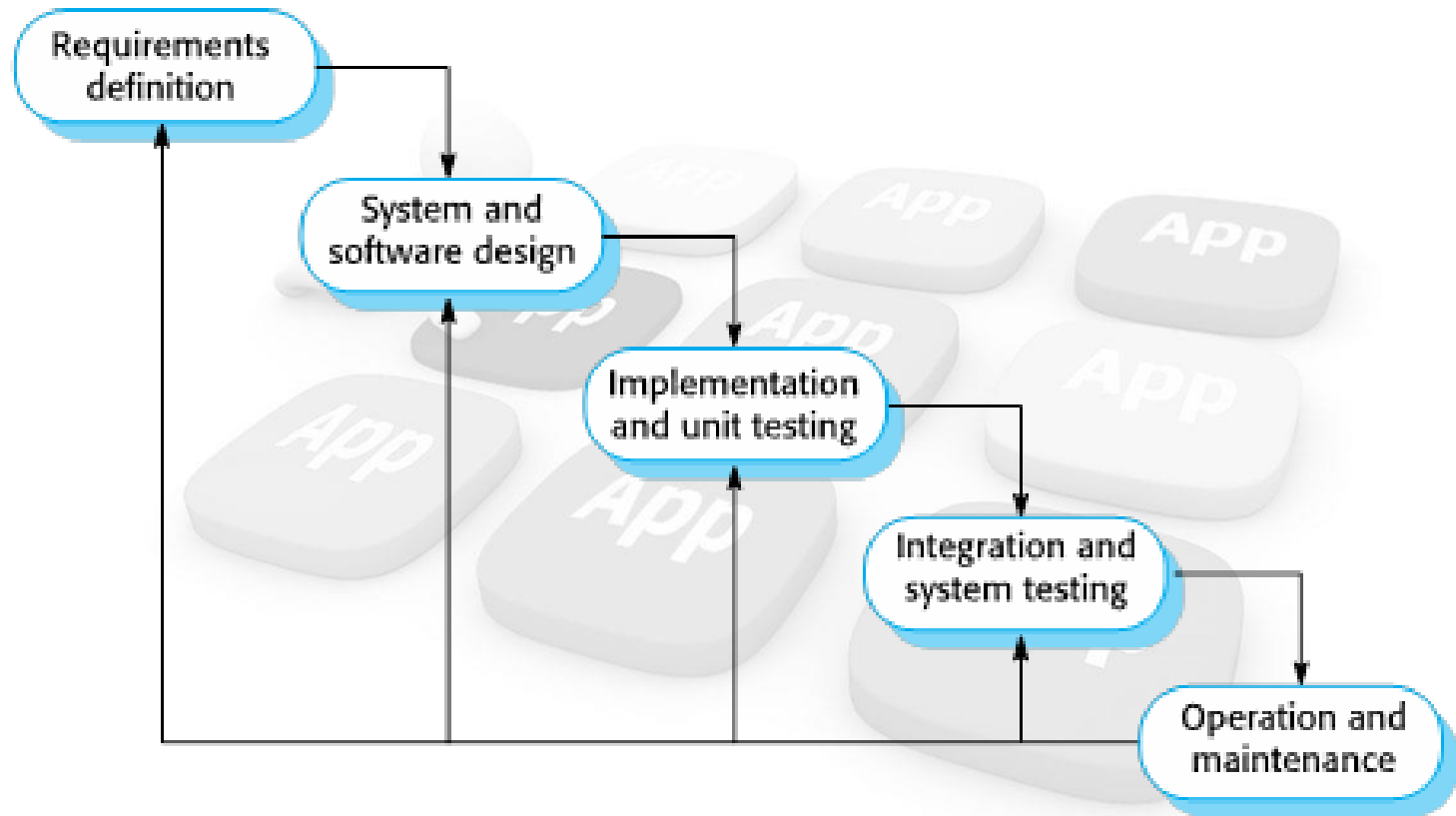
PROJECT PLAN STRUCTURE

- Activity organisation
- Risk analysis
- Hardware and software resource requirements
- Work breakdown
- Project schedule
- Monitoring and reporting mechanisms

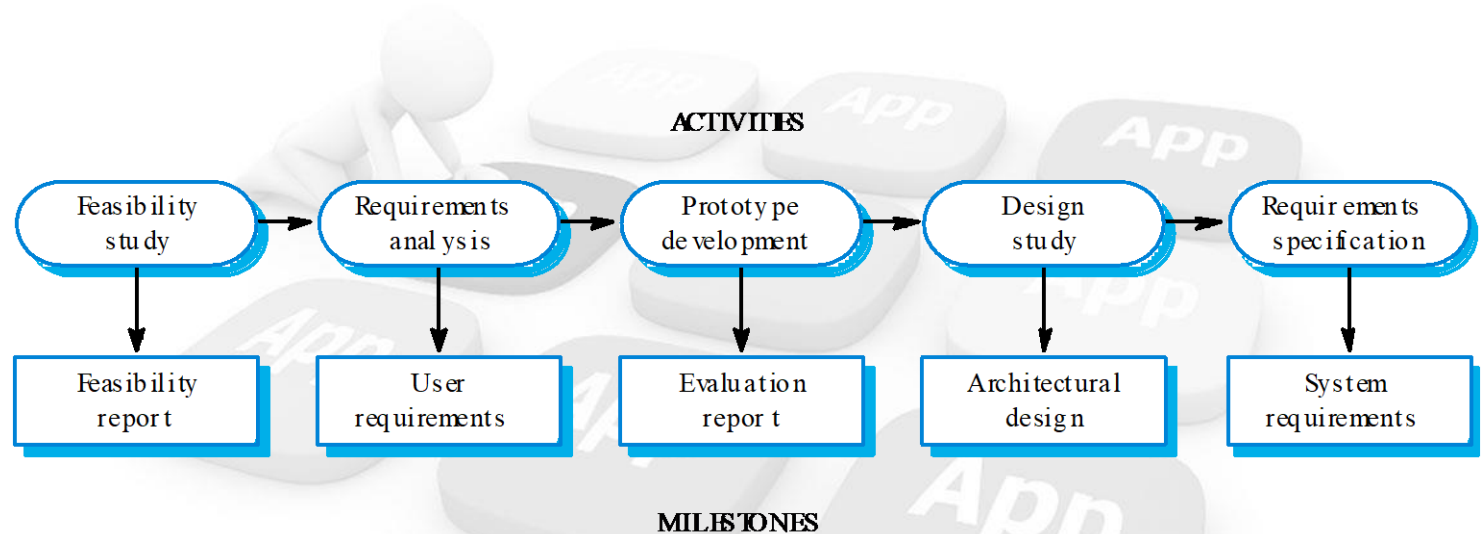
ACTIVITY ORGANIZATION

- Activities in a project should be organised to produce tangible outputs for management to judge progress
- Milestones are the end-point of a process activity
- Deliverables are project results delivered to customers
- The waterfall process allows for the straightforward definition of progress milestones

DELIVERABLES IN THE SDLC



MILESTONES IN THE RE PROCES



RISK MANAGEMENT

- Risk management is concerned with identifying risks and drawing up plans to minimise their effect on a project
- A risk is a probability that some adverse circumstance will occur
 - Project risks affect schedule or resources
 - Product risks affect the quality or performance of the software being developed
 - Business risks affect the organisation developing or procuring the software

SOFTWARE RISKS

Risk	Affects	Description
Staff turnover	Project	Experienced staff will leave the project before it is finished.
Management change	Project	There will be a change of organizational management with different priorities.
Hardware unavailability	Project	Hardware that is essential for the project will not be delivered on schedule.
Requirements change	Project and product	There will be a larger number of changes to the requirements than anticipated.
Specification delays	Project and product	Specifications of essential interfaces are not available on schedule.
Size underestimate	Project and product	The size of the system has been underestimated.
CASE tool underperformance	Product	CASE tools, which support the project, do not perform as anticipated.
Technology change	Business	The underlying technology on which the system is built is superseded by new technology.
Product competition	Business	A competitive product is marketed before the system is completed.

THE RISKS MANAGEMENT PROCESS

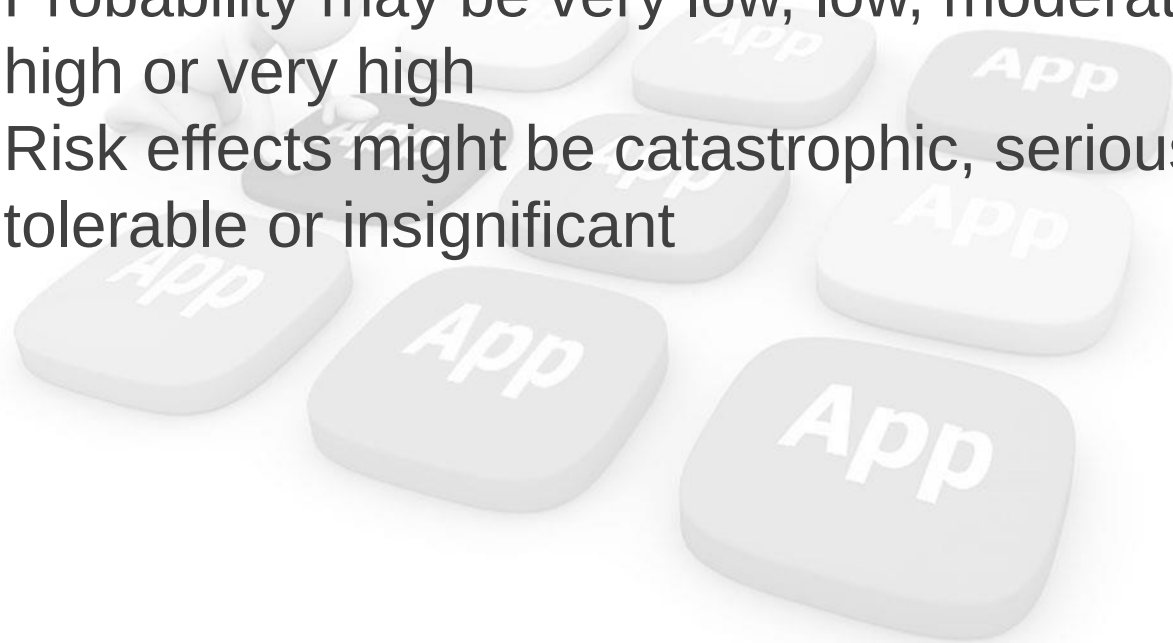
- Risk identification
 - Identify project, product and business risks
- Risk analysis
 - Assess the likelihood and consequences of these risks
- Risk planning
 - Draw up plans to avoid or minimise the effects of the risk
- Risk monitoring
 - Monitor the risks throughout the project

RISKS & RISK TYPES

Risk type	Possible risks
Technology	The database used in the system cannot process as many transactions per second as expected. (1) Reusable software components contain defects that mean they cannot be reused as planned. (2)
People	It is impossible to recruit staff with the skills required. (3) Key staff are ill and unavailable at critical times. (4) Required training for staff is not available. (5)
Organizational	The organization is restructured so that different management are responsible for the project. (6) Organizational financial problems force reductions in the project budget. (7)
Tools	The code generated by software code generation tools is inefficient. (8) Software tools cannot work together in an integrated way. (9)
Requirements	Changes to requirements that require major design rework are proposed. (10) Customers fail to understand the impact of requirements changes. (11)
Estimation	The time required to develop the software is underestimated. (12) The rate of defect repair is underestimated. (13) The size of the software is underestimated. (14)

RISK ANALYSIS

- Assess probability and seriousness of each risk
- Probability may be very low, low, moderate, high or very high
- Risk effects might be catastrophic, serious, tolerable or insignificant



RISK ANALYSIS 1/2

Risk	Probability	Effects
Organizational financial problems force reductions in the project budget (7).	Low	Catastrophic
It is impossible to recruit staff with the skills required for the project (3).	High	Catastrophic
Key staff are ill at critical times in the project (4).	Moderate	Serious
Faults in reusable software components have to be repaired before these components are reused. (2).	Moderate	Serious
Changes to requirements that require major design rework are proposed (10).	Moderate	Serious
The organization is restructured so that different management are responsible for the project (6).	High	Serious
The database used in the system cannot process as many transactions per second as expected (1).	Moderate	Serious

RISK ANALYSIS 2/2

Risk	Probability	Effects
The time required to develop the software is underestimated (12).	High	Serious
Software tools cannot be integrated (9).	High	Tolerable
Customers fail to understand the impact of requirements changes (11).	Moderate	Tolerable
Required training for staff is not available (5).	Moderate	Tolerable
The rate of defect repair is underestimated (13).	Moderate	Tolerable
The size of the software is underestimated (14).	High	Tolerable
Code generated by code generation tools is inefficient (8).	Moderate	Insignificant

IDENTIFICATION AND ANALYSIS OF RISK

			PROBABILITY				
			Frequent	Likely	Occasional	Seldom	Unlikely
			A	B	C	D	E
Severity	Catastrophic	I	Extremely High				
	Critical	II	High	High			
	Marginal	III		Moderate			
	Negligible	IV		Low			

RISK PLANNING

- Consider each risk and develop a strategy to manage that risk
 - Avoidance strategies
 - The probability that the risk will arise is reduced
 - Minimisation strategies
 - The impact of the risk on the project or product will be reduced
 - Contingency plans
 - If the risk arises, contingency plans are plans to deal with that risk

RISK MANAGEMENT STRATEGIES

Risk	Strategy
Organizational financial problems	Prepare a briefing document for senior management showing how the project is making a very important contribution to the goals of the business and presenting reasons why cuts to the project budget would not be cost-effective.
Recruitment problems	Alert customer to potential difficulties and the possibility of delays; investigate buying-in components.
Staff illness	Reorganize team so that there is more overlap of work and people therefore understand each other's jobs.
Defective components	Replace potentially defective components with bought-in components of known reliability.
Requirements changes	Derive traceability information to assess requirements change impact; maximize information hiding in the design.
Organizational restructuring	Prepare a briefing document for senior management showing how the project is making a very important contribution to the goals of the business.
Database performance	Investigate the possibility of buying a higher-performance database.
Underestimated development time	Investigate buying-in components; investigate use of a program generator.

RISK MONITORING

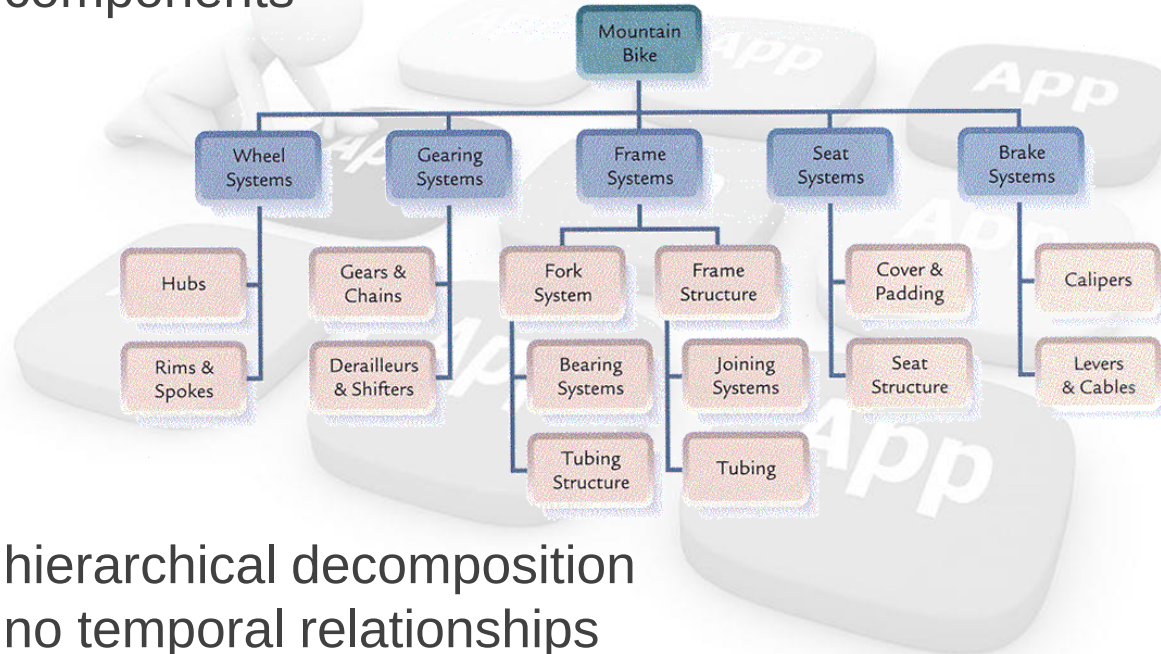
- Assess each identified risks regularly to decide whether or not it is becoming less or more probable
- Also assess whether the effects of the risk have changed
- Each key risk should be discussed at management progress meetings

RISK INDICATORS

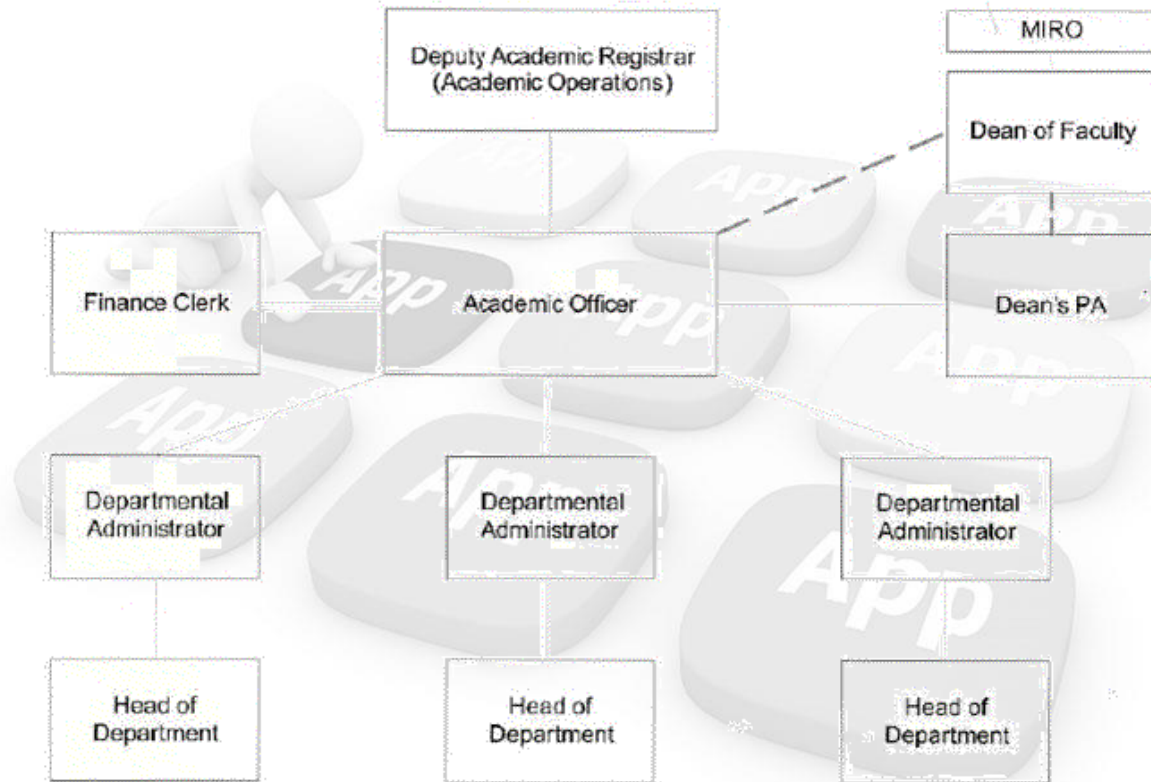
Risk type	Potential indicators
Technology	Late delivery of hardware or support software; many reported technology problems.
People	Poor staff morale; poor relationships amongst team members; high staff turnover.
Organizational	Organizational gossip; lack of action by senior management.
Tools	Reluctance by team members to use tools; complaints about CASE tools; demands for higher-powered workstations.
Requirements	Many requirements change requests; customer complaints.
Estimation	Failure to meet agreed schedule; failure to clear reported defects.

WORK BREAKDOWN

- Work breakdown structure (WBS) as refinement of components



LINCOLN WBS

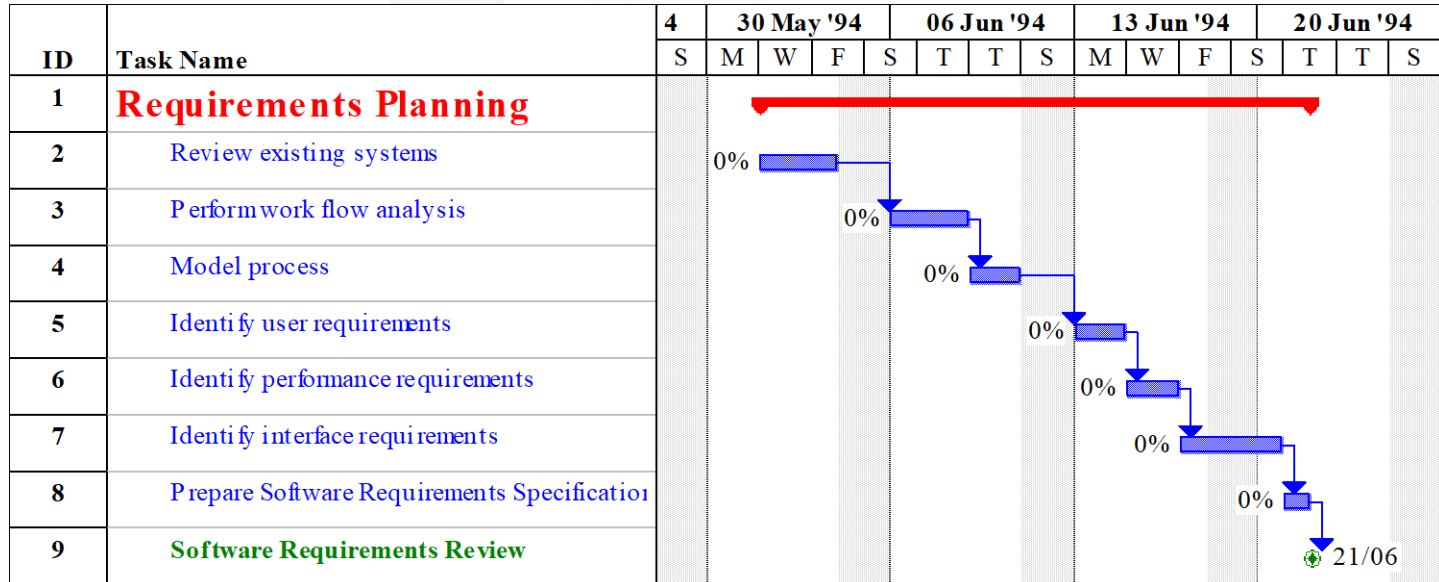


PROJECT SCHEDULING

- Split project into tasks and estimate time and resources required to complete each task
- Organize tasks concurrently to make optimal use of workforce (if possible)
- Minimize task dependencies to avoid delays caused by one task waiting for another to complete
- Dependent on project managers intuition and experience

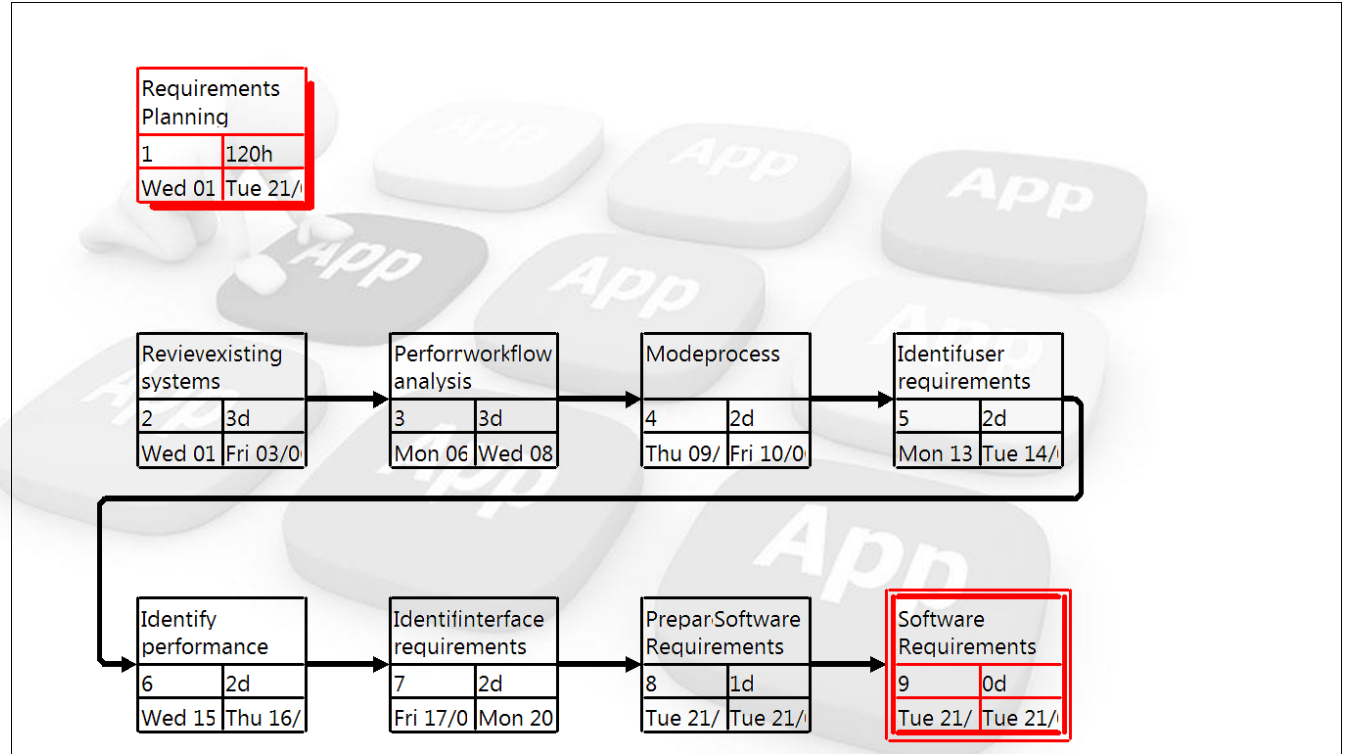
THE REQUIREMENTS PHASE : 27/35

GANTT



THE REQUIREMENTS PHASE : PERT

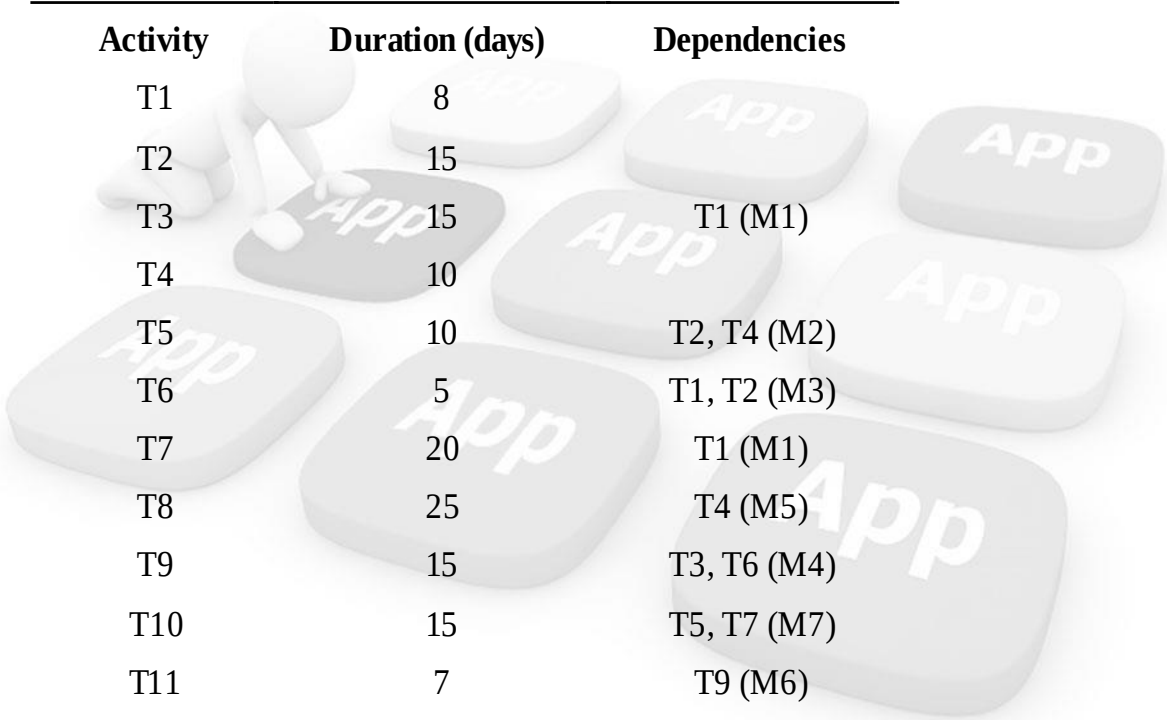
28/35



SCHEDULING PROBLEMS

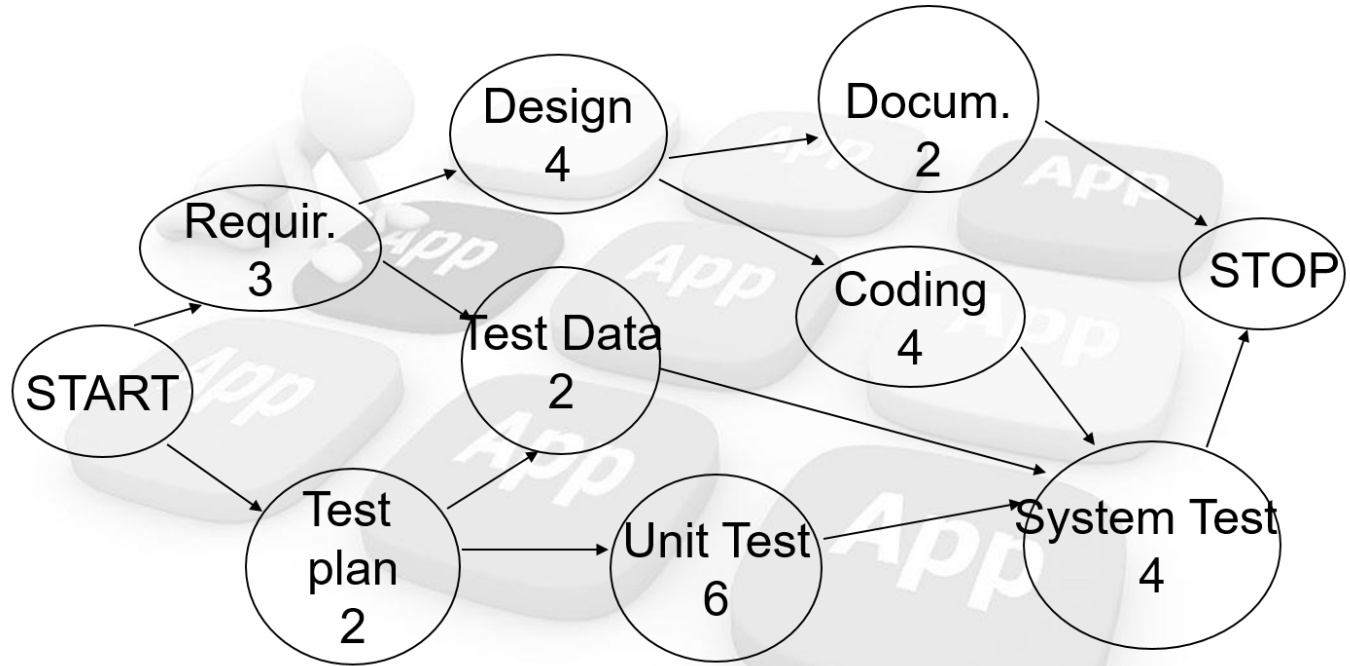
- Estimating the difficulty of problems and hence the cost of developing a solution is hard
- Productivity is not proportional to the number of people working on a task
- Adding people to a late project makes it later because of communication overheads
- The unexpected always happens. Always allow contingency in planning

TASK DURATIONS & DEPENDENCIES



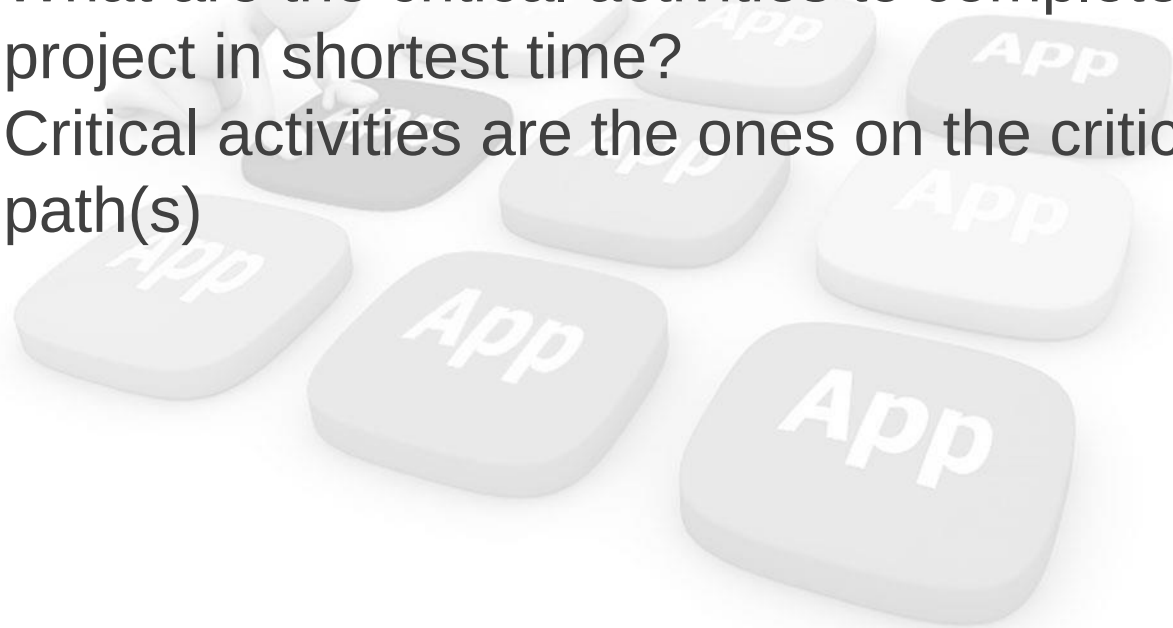
Activity	Duration (days)	Dependencies
T1	8	
T2	15	
T3	15	T1 (M1)
T4	10	
T5	10	T2, T4 (M2)
T6	5	T1, T2 (M3)
T7	20	T1 (M1)
T8	25	T4 (M5)
T9	15	T3, T6 (M4)
T10	15	T5, T7 (M7)
T11	7	T9 (M6)
T12	10	T11 (M8)

PERT CONCURRENT ACTIVITIES

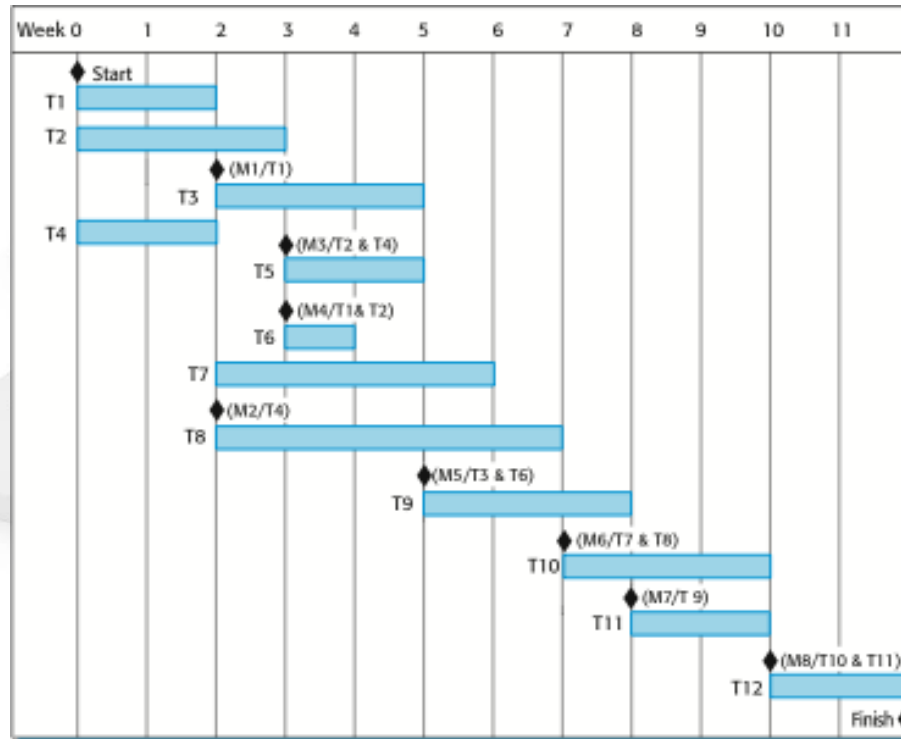


CRITICAL PATH ANALYSIS

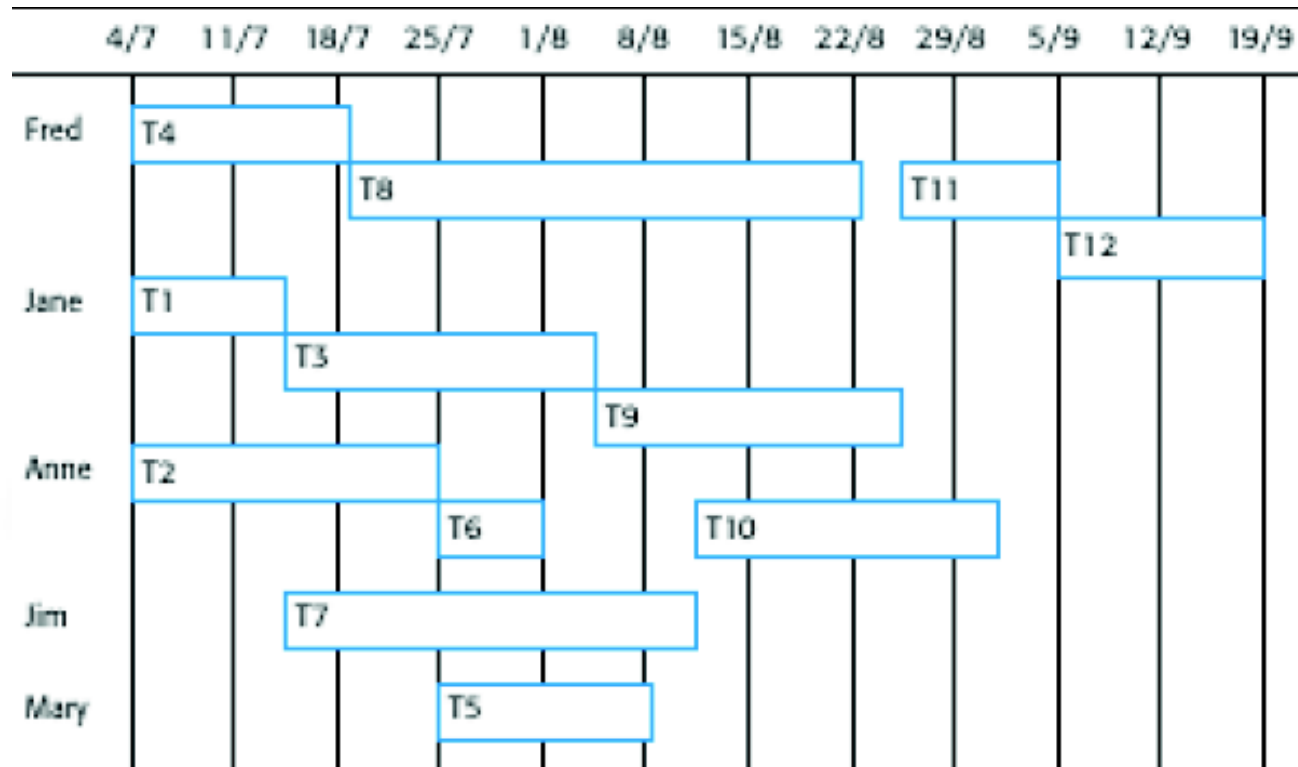
- What is shortest time to complete the project?
- What are the critical activities to complete the project in shortest time?
- Critical activities are the ones on the critical path(s)



ACTIVITY TIMELINE



STAFF ALLOCATION



THANK YOU...



اَوْنُو تِكْنُوْلُوجِي مَارَا
UNIVERSITI
TEKNOLOGI
MARA